HEBREW · DYSLEXIA · ACCOMMODATIONS · ISRAEL

Hebrew exam reading, *redesigned.*

A reader for dyslexic Israeli students that's not human — but reads like one. The case for tirvi: the gap, the audience, and the architecture that closes it.

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01 / PROBLEM

Hebrew exam reading is broken.

Israeli students with dyslexia are entitled to a human reader during exams. When the human is replaced by software, today's software fails them — not because Hebrew TTS is bad, but because no one wired Hebrew NLP into a reader that understands an exam.

Three reasons today's tools don't work

- **Pronunciation guesses.** Hebrew is written without vowels. Generic TTS guesses, gets homographs wrong, and a dyslexic ear can't recover from the error.
- **Flat reading.** Question stems, answer choices, table cells, math — all read top-to-bottom as one stream. A human reader re-reads the choices on demand. Software doesn't.
- **No accommodation-grade UX.** No replay-this-sentence, no slow-this-word, no jump-to-question, no word-sync highlight. Public criticism of Israel's 2024 computerized-reading pilot says it directly.

> *"Cold, rigid, not adjustable in real time."*
>
> PUBLIC CRITICISM — ISRAELI MOE 2024 COMPUTERIZED-READING PILOT FOR BAGRUT MATH, REPLACING HUMAN READERS FOR STUDENTS WITH DYSCALCULIA / DYSGRAPHIA. YNET, 2024.

54%

of Israeli high-schoolers received Bagrut accommodations in 2021 — up from 35% in 2011. Level-1 accommodations alone jumped 8% → 40% (2016–2021).

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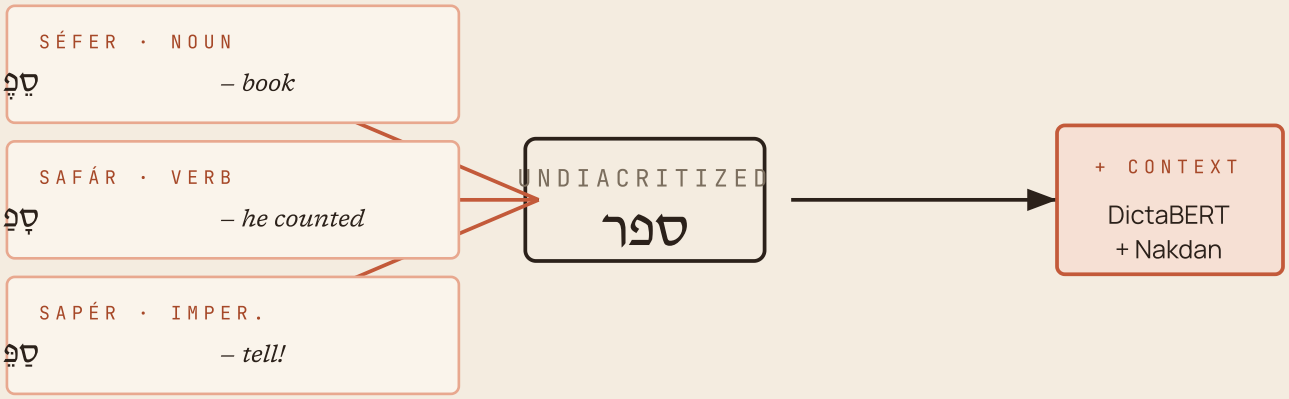
01.1 / WHY HEBREW IS HARD

Three things English readers never face.

English TTS works because English on the page already encodes the pronunciation. Hebrew doesn't. The reader has to think before it can speak.



01 · NO VOWELS

Same letters, three readings.

Hebrew omits ניקוד. ספר is book / counted / tell. Only morphology + context can choose. Generic TTS picks one and ships it.

02 · ACRONYMS

Heavy density, idiomatic expansion.

ד״ר → doctor, עמ׳ → page, מס׳ → number. Generic TTS spells them letter-by-letter. Hebrew exam content is full of them.

03 · MIXED DIRECTION

RTL meets LTR, mid-sentence.

Bagrut math and English mix Hebrew with English, numbers, formulas. Google's SSML `<lang>` tag returns silence on Hebrew. One sentence, three scripts, no clean handover.

02 / MARKET SIZE

The accommodation cohort, by the numbers.

Israel formally classifies more dyslexic students than almost any peer system, and the share is still rising.

~500K

Israeli students estimated to have dyslexia — about three to five in every class of thirty.

ISRAEL NATIONAL NEWS

54%

of Israeli high-schoolers received Bagrut accommodations in 2021 — vs. ~15% international LD prevalence.

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+19pp

growth in the accommodation rate over a decade (2011 → 2021), driven by Level-1 accommodations.

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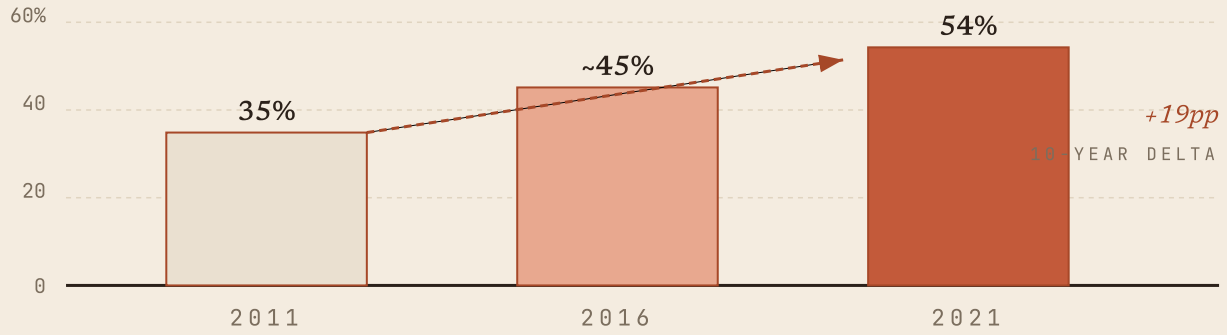
Bagrut accommodations rate, Israeli high-schoolers



SOURCE · TAUB CENTER 2024 · BAGRUT EXAM ACCOMMODATIONS

02.1 / EXISTING TOOLS

Generic readers, generic results.

No commercial product today combines Hebrew OCR, Hebrew NLP, and Hebrew TTS into an exam-shaped reader. The building blocks exist. No one has wired them together.

TOOL	HEBREW TTS	EXAM STRUCTURE	DIACRITIZATION	WORD-SYNC	VERDICT
ElevenLabs (v3, Flash v2.5)	●	○	○	●	Generic API
Speechify	●	○	○	●	Generic article reader
NaturalReader	●	○	○	◐	Generic doc TTS
Voice Dream Scanner	○	◐	○	●	Hebrew not supported
Kurzweil 3000	○	●	○	●	No Hebrew voice
MS Immersive Reader	◐	◐	○	◐	RTL layout issues
Read&Write (Texthelp)	○	◐	○	●	No first-class Hebrew
Israeli MoE 2024 pilot	●	○	◐	○	“Cold, rigid, not adjustable”
tirvi	●	●	●	●	Built for Hebrew exams

● FULL SUPPORT ◐ PARTIAL / LIMITED ○ NONE

The building blocks – DictaBERT, Dicta-Nakdan, Phonikud, Google he-IL TTS – all exist. No one has assembled them into a vertically integrated reader for exam content. — TIRVI RESEARCH SYNTHESIS · APRIL 2026

03 / SOLUTION

Don't add TTS to OCR. *Add a brain.*

The defensible layer is the middle stage — Hebrew morphology, disambiguation, diacritization, and reading-plan shaping. OCR and TTS are commodities behind adapters.

01 · INGEST	02 · INTERPRET — THE MOAT	03 · SHAPE	04 · VOICE	05 · LISTEN
Hebrew OCR	**Hebrew NLP**	**Reading Plan**	**Hebrew TTS**	**Player**
Tesseract `heb` default, Document AI fallback, DeepSeek-OCR pilot. Structured blocks + bboxes, not flat text.	DictaBERT for morphology & disambiguation. Dicta-Nakdan for ניקוד. Phonikud for stress + IPA G2P.	Per-block JSON: tokens, lemma, POS, pronunciation hints, SSML cues, structural type (question / answer / table).	Google Wavenet for word-sync; Chirp 3 HD for continuous; Azure as alternate. Cached by content hash.	Side-by-side viewer, word-sync highlight, per-block playback, repeat sentence, font-size, WCAG 2.2 AA.

The three principles that decide it

- **The reading plan is the product.** 3× as much code in the interpretation layer as in adapters.
- **Adapters return rich result objects, not bytes.** Timepoints, bboxes, confidences survive the port boundary.
- **Cache by content hash, share across users.** The single biggest cost lever and the only way the $0.02 / page target survives premium voices.

Why now

- **2024 MoE pilot** proved the policy direction (digital readers) and the UX gap (cold, rigid).
- **Dicta lineage** (Bar-Ilan) shipped DictaBERT 2.0 in 2024 — SOTA on UD-Hebrew.
- **Phonikud** (June 2025) gives real-time Hebrew G2P with stress & vocal shva — the missing layer.
- **Google Chirp 3 HD he-IL** shipped in 2025 — premium voice quality at last.

03.1 / THE MOAT — HEBREW READING PLAN

Five micro-stages between text and audio.

This is where context-aware Hebrew reading happens — morphological disambiguation, pronunciation prediction, exam-domain adaptation. The academic framing matches the engineering one.

01 · TOKENIZE

Morphology

DictaBERT-large-joint — segmentation, POS, lemma, dependency. SOTA on UD-Hebrew 2024–25.

02 · DISAMBIGUATE

Context

Sentence context picks the right reading for homographs. ספר → verb or noun?

03 · DIACRITIZE

Add ניקוד

Dicta-Nakdan adds vowels for ambiguous tokens. ~86.86% word-level accuracy.

04 · G2P

Phonemes

Phonikud (June 2025) outputs IPA with stress and vocal shva. Real-time, plug into any TTS.

05 · SHAPE

SSML

Breaks between answers, emphasis on question numbers, language switch on English spans, mark for word-sync.

Worked example — one Bagrut question stem

INPUT

ספר את הסיפור במילים שלך.

DISAMBIGUATED

סַפֵּר אֶת הַסִּיפּוּר בְּמִלִּים שֶׁלָּךְ. — *"tell the story in your own words" (verb sapér, not noun séfer)*

SSML

```
<mark name="q4-w0"/><emphasis level="moderate">סַפֵּר</emphasis>
אֶת הַסִּיפּוּר <break time="200ms"/> בְּמִלִּים שֶׁלָּךְ.
```

SPOKEN

*"sa**pér** et ha-sipúr be-milím shelákh"*

04 / TARGETS & PLAN

Three numbers, sixteen weeks.

≥90%

word-pronunciation accuracy on the tirvi-Hebrew-Exam Benchmark v0.

VALIDATED BY BLIND MOS · N=10 DYSLEXIC TEENS

≤30s

time to first audio, p50, on a 5-page scanned exam.

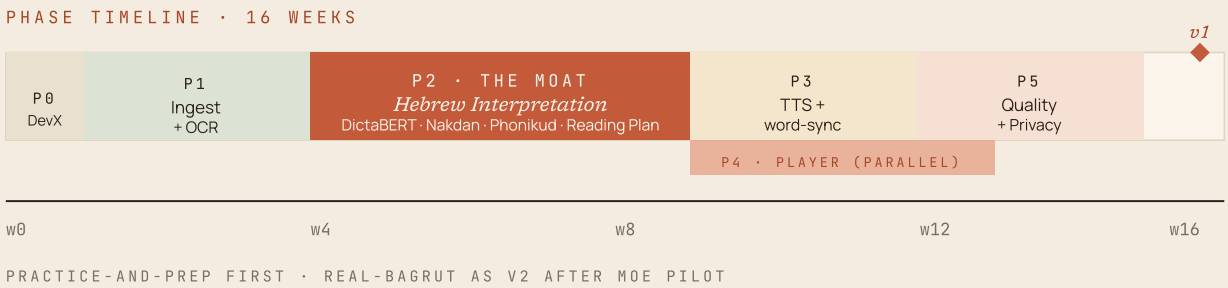
CLOUD RUN MIN-INSTANCES=1 · CPU-BOOST

≤$0.02

amortized cost per processed page across a 30-day window.

WAVENET + CONTENT-HASH CACHE > 25%

Sixteen-week MVP — 11 epics, six phases, practice-mode framing



Out of MVP scope, deferred to v1.1 / v2: handwriting OCR, user accounts (move to small Cloud SQL), real-time WebSocket status, custom voices, math/SRE Hebrew localization, MoE pilot toward in-Bagrut use, mobile-native apps. **Privacy posture:** 24-hour default TTL, DPIA + parental-consent ≥14, upload attestation against third-party copyright.

END MATTER

Where this brief goes next.

Three concrete moves on the desk: a benchmark, an ADR, and an outreach email.

DOCUMENT	tirvi · Market Brief · v1.0
SOURCE SYNTHESIS	docs/research/tirvi-validation-and-mvp-scoping.md
REPOSITORY	VSProjects/tirvi
OPEN ADRS	10 queued · TTS routing, NLP backbone, MVP framing, TTL, OCR primary, …
ACADEMIC CANDIDATES	Bar-Ilan ONLP Lab (Tsarfaty) · Dicta · HUJI adiyoss-lab
FIRST THREE MOVES	1. Stand up tirvi-bench v0 · 2. Run `@design-pipeline` on ADR-006 (MVP framing) · 3. Send the Bar-Ilan / Dicta / HUJI outreach

TIRVI · APRIL 2026 · V1.0 JISHUTECH.IO